Newton’s Third Law

Review: Newton's Laws

- Newton's laws of motion:

  1. An object at rest stays at rest, and an object moving with constant velocity keeps moving with constant velocity, unless a net force acts on it.

  2. At any instant, the net force on an object equals $\vec{F}_{\text{net}} = m\vec{a}$

  3. (Topic of this week)

Review: Force Models

- Mechanical equilibrium model
  - Object at rest or moving with constant velocity
  - $\vec{F}_{\text{net}} = m\vec{a} = 0$
- Constant force model
  - All forces are constant
  - $\vec{F}_{\text{net}} = m\vec{a} =$ constant, so $\vec{a} = \text{constant}$
  - Use kinematic equations

Review: Using Newton's Second Law

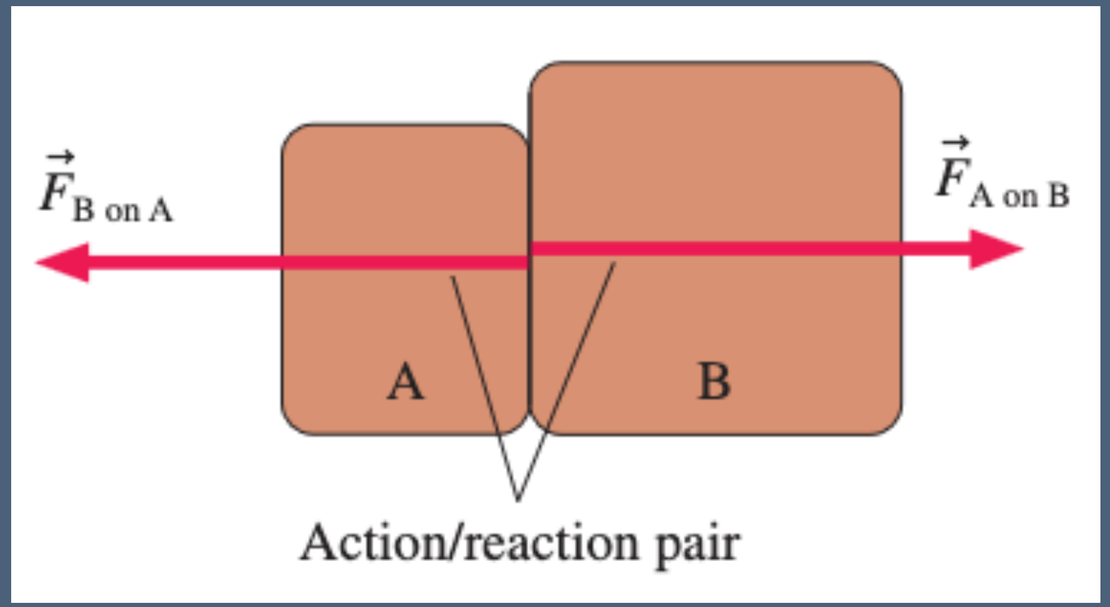
- Identify the physics: equilibrium or constant force? Free fall, projectile motion, circular motion, static or kinetic friction?
- Draw a picture, choose a coordinate system, define symbols, and identify what the problem asks for
- Identify every force acting on the object at that instant
- Draw a free-body diagram
- Then either use:
  - $\vec{F}_{\text{net}}$ to find $\vec{a}$, then use kinematics;
  - kinematics to find $\vec{a}$, then $\vec{F}_{\text{net}} = m\vec{a}$ to find unknown forces

Interacting Objects

- So far: forces acting on a single object
- Now: two or more objects that interact with each other
- Goal: figure out how the objects move
- Often we care about the motion of only some of the objects

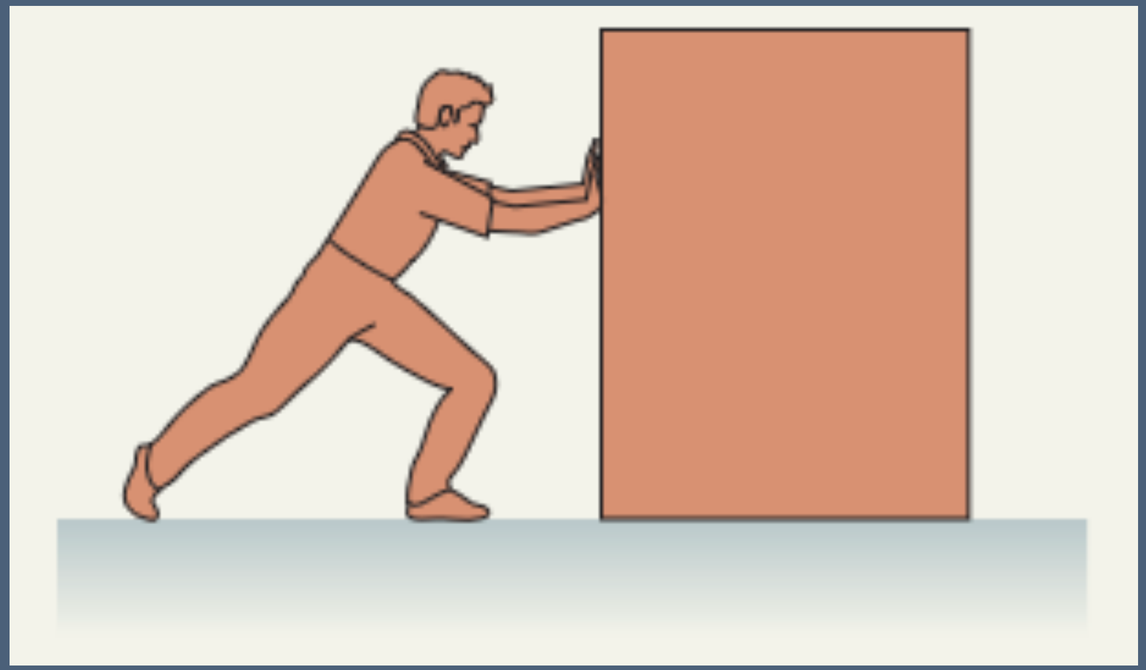
Newton's Third Law

- Interaction: the mutual influence of two objects on each other
- In this course, every force we use is one half of an action/ reaction pair (or pairwise forces)
- Newton's third law: the two forces in a pair
  - have the same magnitude
  - point in opposite directions
  - act on different objects
  - are the same type of force
  - $\vec{F}_{A \text{ on } B} = -\vec{F}_{B \text{ on } A}$
- This holds for contact forces and long-range forces



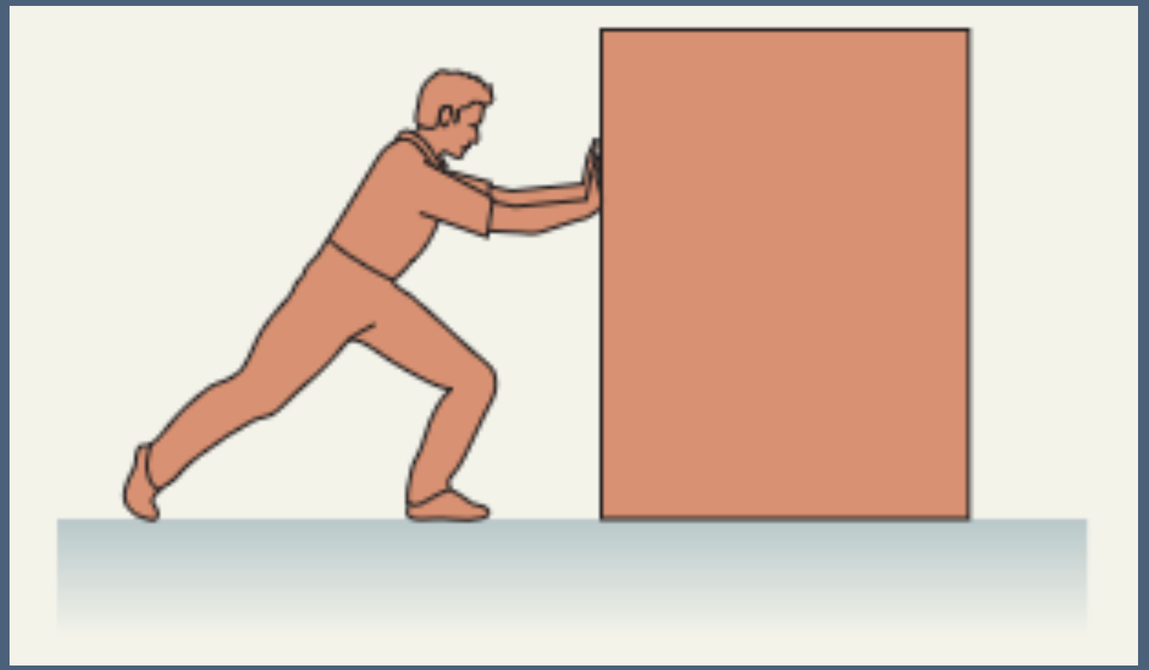
Example

- Consider a person pushing a crate across a rough surface. What are all the interactions?

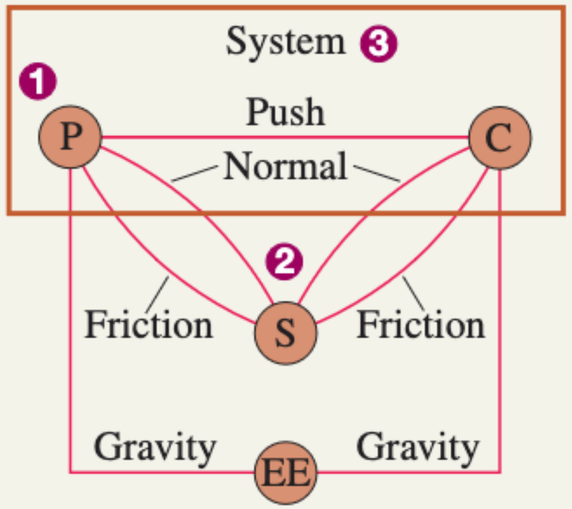


Example

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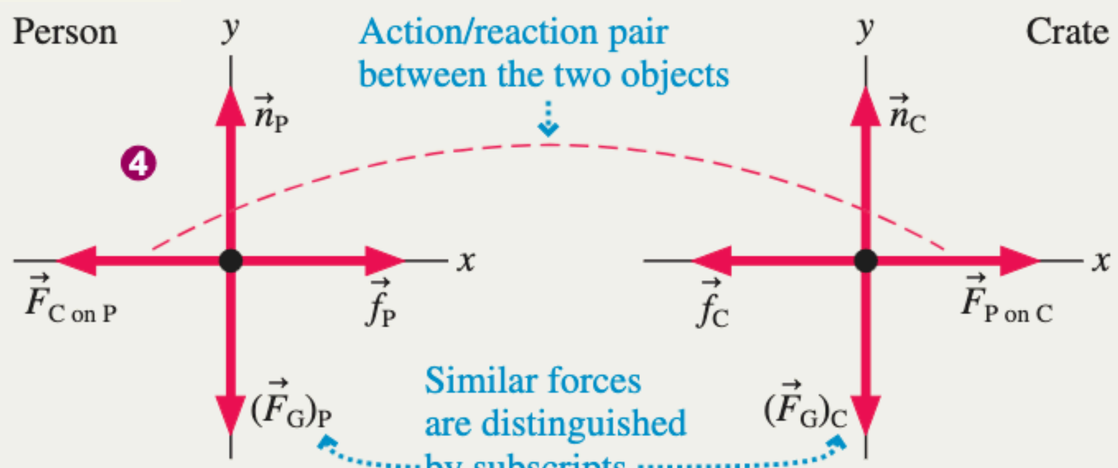


The interaction diagram.



P = Person
C = Crate
S = Surface
EE = Entire earth

Free-body diagrams of the person and the crate.



Common Trap: Is it a Pair?

- A book rests on a table. Its weight (down) and the normal force (up) are equal and opposite. Are they action/reaction pair?
- No. Both act on the book, and they are different types of force.
- The real partners:
  - Earth pulls down on the book —> the book pulls up on the Earth (gravity)
  - The table pushes up on the book —> the book pushes down on the table (contact)
- Weight and normal force are equal here because the second law $\vec{F}_{\text{net}} = 0$, not the third law!

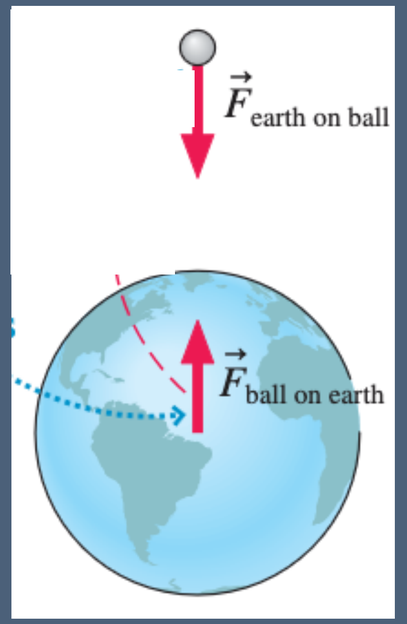
Equal Forces, Unequal Effects

- The forces in a pair are equal, but the accelerations they cause are not, because $\vec{a} = \dfrac{\vec{F}_{\text{net}}}{m}$

- Example: you drop a $1\,\text{kg}$ ball. The ball and earth pull on each other with equal and opposite forces. Why doesn't the earth move?

$$\vec{F}_{\text{earth on ball}} = -m_B g\hat{j} \longrightarrow \vec{a}_{\text{ball}} = \frac{\vec{F}_{\text{earth on ball}}}{m_B} = -g\hat{j}$$

$$\vec{F}_{\text{ball on earth}} = -\vec{F}_{\text{earth on ball}} = m_b g\hat{j} \longrightarrow \vec{a}_{\text{earth}} = \frac{\vec{F}_{\text{ball on earth}}}{m_E} = \frac{m_B}{m_E} g\hat{j} = \frac{1\,\text{kg}}{6\times 10^{24}\,\text{kg}} g \approx 2\times 10^{-24}\text{m/s}^2$$



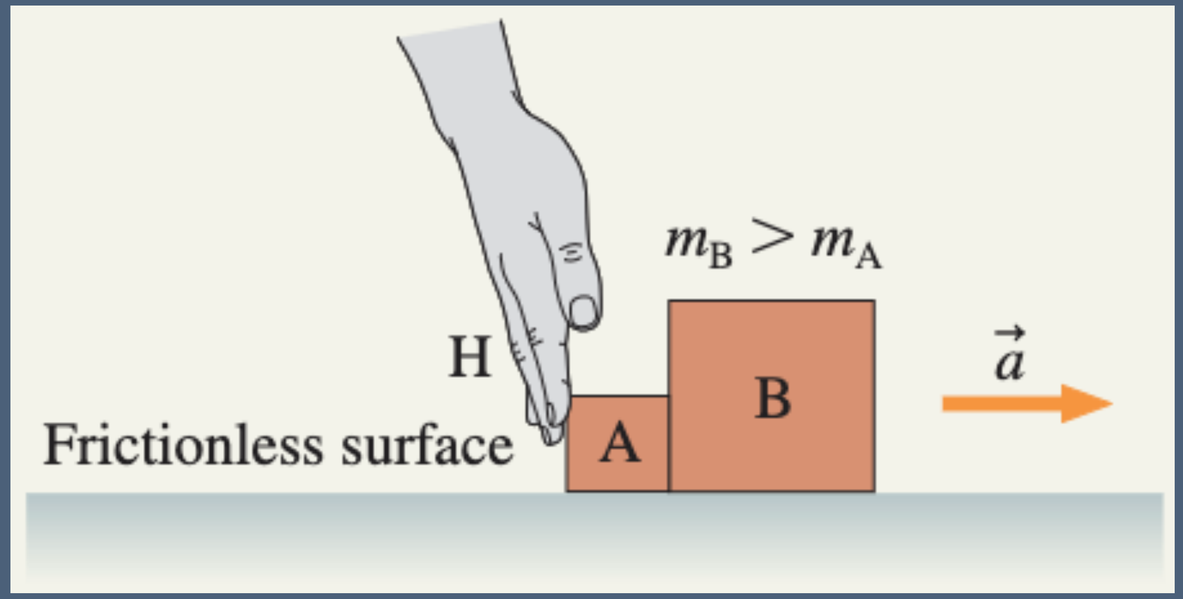
- When two objects of very different mass interact, the lighter one does essentially all the accelerating

System and Environment

- **System**: the objects whose motion we want to analyze
- **Environment**: everything outside the system
- **External forces**: forces acting on the system caused by objects in the environment
  - When doing problems, we ignore the pair partners caused by objects in the system acting on the environment
- **Internal forces**: forces between objects included in the system
  - When doing problems, internal forces cancel in the net force of the system

Example

- A hand pushes box A and B to the right across a frictionless table with constant acceleration $\vec{a}$. The mass of B is greater than the mass of A.
- What is the system? What is the environment?
- Draw a free body diagram of the horizontal forces for A and B
- What are the internal and external forces?
- Order the net forces from largest to smallest.



Example

- A hand pushes box A and B to the right across a frictionless table with constant acceleration $\vec{a}$. The mass of B is greater than the mass of A.
- What is the system? What is the environment?

The system is block A and B; the environment is the hand and the surface.

- Draw a free body diagram of the horizontal forces for A and B
- What are the internal and external forces?



F_{AB} and F_{BA} are internal. F_{HA} is external

- Order the forces from largest to smallest.

$F_{\text{net}-\text{system},x} = m_{\text{system}}a_{\text{system}} = (m_A + m_B)a = F_{HA} - F_{BA} + F_{AB} = F_{HA}$

$F_{\text{net},A} = m_A a = F_{HA} - F_{BA}$, $F_{\text{net},B} = m_B a = F_{AB}$

$F_{\text{net}-\text{system},x} = (m_A + m_B)a = F_{HA} \rightarrow a = \frac{F_{HA}}{m_A + m_B}$, so $F_{\text{net},B} = m_B a = \frac{m_B}{m_A + m_B} F_{HA}$ and $F_{\text{net},A} = m_A a = \frac{m_A}{m_A + m_B} F_{HA}$

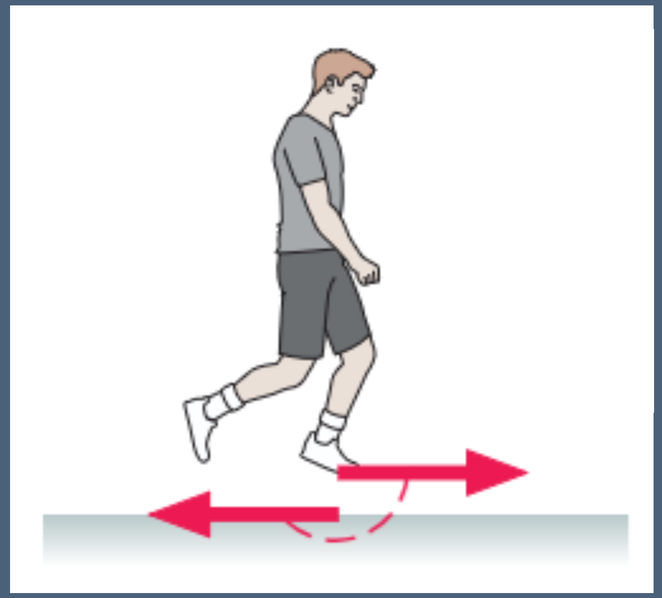
$F_{\text{net}-\text{system},x} > F_{\text{net},B} > F_{\text{net},A}$

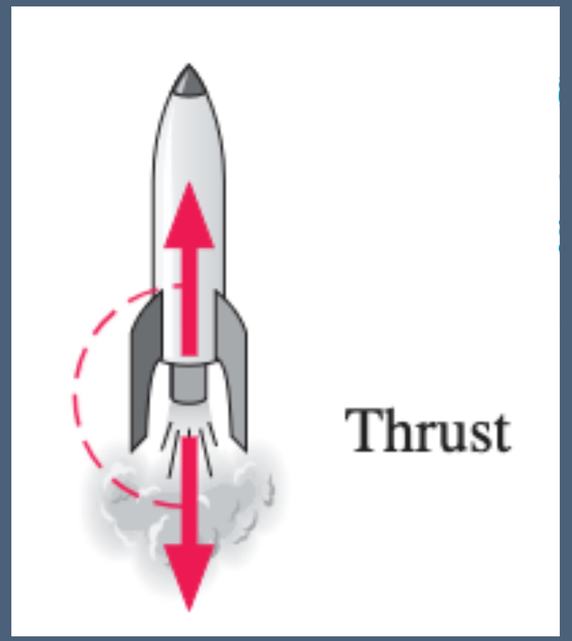
Takeaway

- **Newton's second law works for any object or any collection of objects you choose**
  - The choice decides which forces count as external
  - Force internal to the chosen system cancel in $\vec{F}_{\text{net}}$
- For the system: $\vec{F}_{\text{net}-\text{system}} = m_{\text{system}}\vec{a}_{\text{system}}$
  - Net forces on individual objects add up to the net force on the system: $\vec{F}_{\text{net}-\text{system}} = \vec{F}_{\text{net},A} + \vec{F}_{\text{net},B} + \dots$ (internal pairs cancel in the sum)
  - $\vec{a}_{\text{system}}$ is shared acceleration when the objects move together (acceleration constraint)
- For an object: $\vec{F}_{\text{net}-\text{object}} = m_{\text{object}}\vec{a}_{\text{object}}$
  - Internal pairwise forces always cancel because both halves act on objects inside the system and they are equal in magnitude but opposite in direction.
  - The partners to external forces act on the environment so they don't show up in $\vec{F}_{\text{net}}$

Propulsion

- An object can’t push itself forward. It has to push on something else, and that something pushes back.
- Muscles, engines, and propellant let the object do the pushing, but are internal forces so they can’t accelerate the object themselves
- The force that actually accelerates the system comes from the environment, as a pairwise partner.



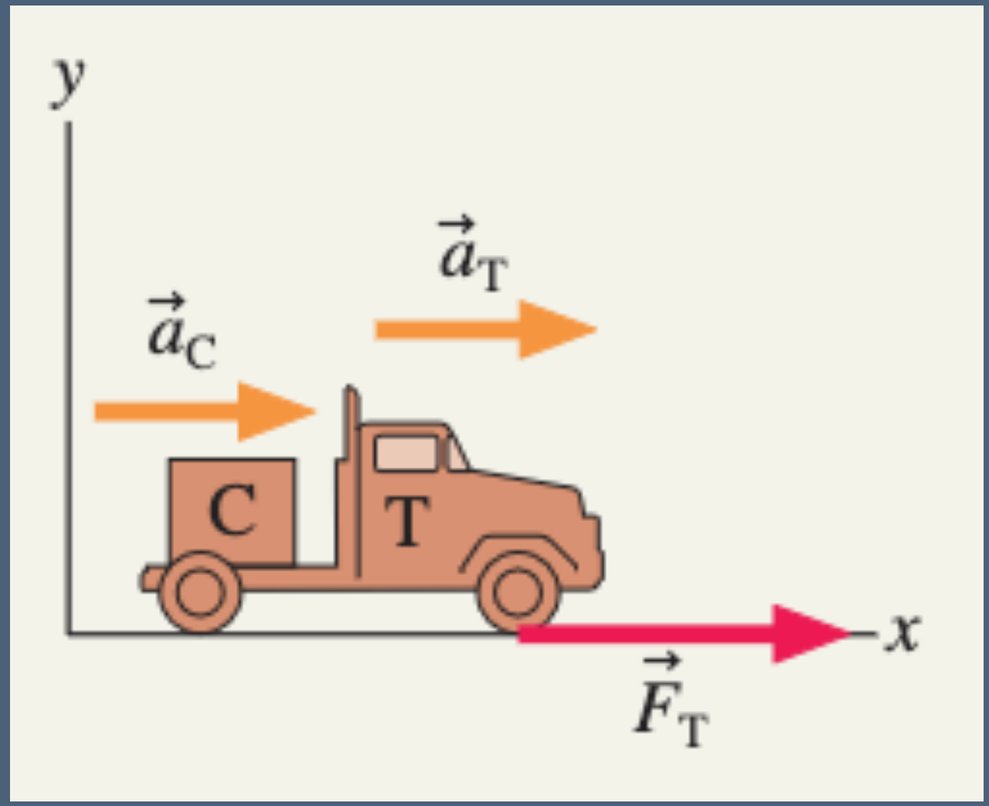


Closer look at walking

- Internal forces:
  - muscle pulls on bone, bone pulls back on muscle
  - upper leg pushes on lower leg, lower leg pushes back
  - Both halves of each pair act on part of the person, so they cancel in $\overrightarrow{F}_{\text{net}-\text{system}}$.
  - The job of these forces is to move the foot relative to the body, driving it backwards against the ground
- Foot pushing backwards on ground (this acts on the environment so not in $\overrightarrow{F}_{\text{net}-\text{system}}$)
- External forces:
  - The pairwise partner from the ground pushing forward on the foot (static friction)
  - This is what accelerates the person

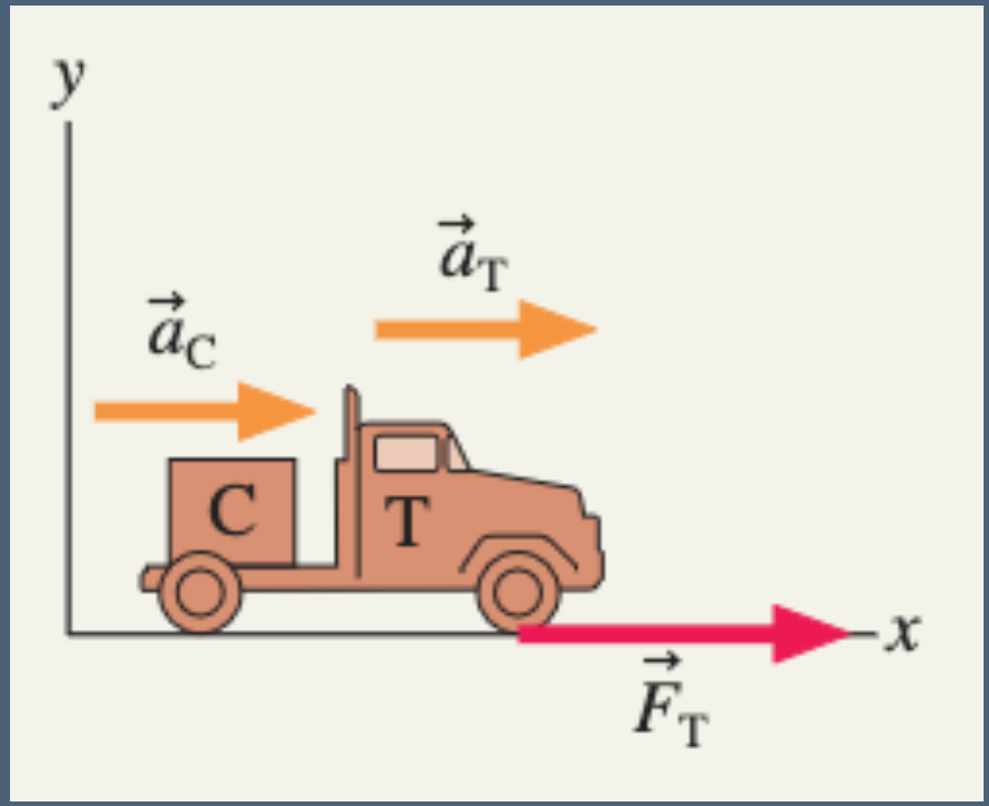
Example

- You and a friend have just loaded a $200\,\mathrm{kg}$ crate filled with priceless art objects into the back of a $2000\,\mathrm{kg}$ truck. As you press down on the accelerator, force $\vec{F}$ surface on truck propels the truck forward. To keep things simple, call this $\vec{F}_T$. What is the maximum magnitude $\vec{F}_T$ can have without the crate sliding? The static and kinetic coefficients of friction between the crate and the bed of the truck are 0.80 and 0.30. Rolling friction of the truck is negligible.



Example

- $F_{T,\max} = \mu_s g(m_T + m_C) = 17000\ \text{N}$



Ropes: The Massless String Approximation

- A rope can only pull, never push.
- Real rope has mass m_S.
- If rope accelerates: $F_{\text{net},x} = T_{B \text{ on } S} - T_{A \text{ on } S} = m_S a_x$, so the tension at the two ends are different!

Ropes: The Massless String Approximation

- If rope's mass much smaller than the objects' masses, use the massless string approximation: $m_S \to 0$, so $F_{\text{net},x} \to 0$ and
- Now tension is the same at every point along a massless string: $T_{B \text{ on } S} = T_{A \text{ on } S}$
- From this, we can leave the string out of system and treat $\vec{F}_{A \text{ on } B}$ and $\vec{F}_{B \text{ on } A}$ like an action/reaction pair
  - **A massless string just passes the force along without changing its size**
- This isn't literally true, since A and B each interact with the string, not with each other. and $\vec{F}_{A \text{ on } B}$ and $\vec{F}_{B \text{ on } A}$ might not have opposite direction!

Pulleys

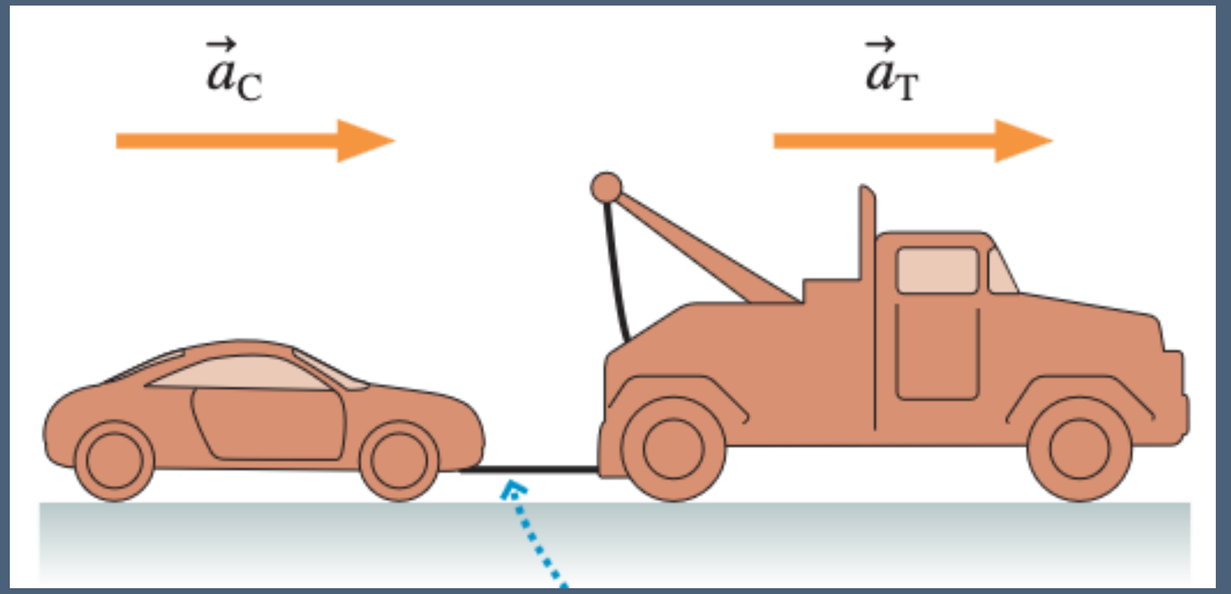
- Assume the pulley is massless and turns without friction, so **no force is needed to turn it**
- Then the tension in a massless string is the same on both sides of the pulley
- Each object feels a tension of the same size T, pointing along the string away from the object
- The pulley only changes the **direction** of the tension, not its **magnitude**
- We leave the string and pulley out of the free-body diagrams.
- These forces act like an action/reaction pair in magnitude only. They don't point in opposite directions

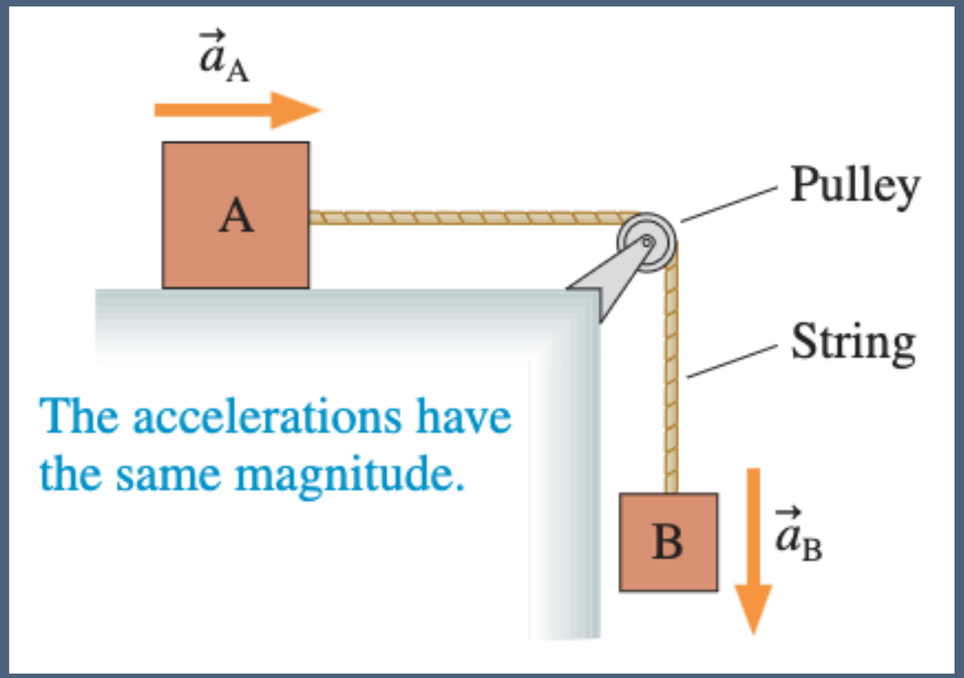
Review

- Newton's second law, $\vec{F}_{\text{net}} = m_{\text{net}}\vec{a}$ applies to any object or collection of objects
- Newton's third law: action/reaction pairs are equal in magnitude but opposite in direction and apply to different objects
  - All forces in this class are pairwise forces
- Massless string only transmits the tension force, tension on both ends like action/reaction pair (not actually true)
- Massless, frictionless pulleys only change the direction of the force

Acceleration Constraints

- When objects are connected or stay in contact, their motions are linked
- **Acceleration constraint**: a relationship between accelerations of two or more objects
  - Gives you an extra equation for solving the problem
- Car and two truck move together, $\vec{a}_C = \vec{a}_T$
- Blocks connected by a string over a pulley: same magnitude, different direction, $a_{A,x} = a_{B,y}$





Problem-Solving Strategy: Interacting Objects

1. Prepare
    - Decide which objects are in the system and which are in the environment
    - Make simplifying assumptions (massless string, frictionless pull, etc.)
2. Picture
    - Sketch the situation (you may give each object its own coordinate system)
    - Define symbols, write the acceleration constraints, and identify what you need to find
3. Free-body diagrams
    - Draw a separate diagram for each object, showing only forces acting on that object
    - Connect action/reaction pairs with dashed lines
4. Solve
    - Write Newton's second law for each object
    - Set the magnitudes of action/reaction pairs equal
    - Add the constraints, friction model, and any other given information
    - Solve for the acceleration, then use kinematics for velocities and positions